

PRELAB ASSIGNMENT – Taylor Series and Trigonometric Functions

1.

The Taylor Series is a method to approximate complicated functions using simple polynomials just like x , x^2 , x^3 . Instead of calculating a complex function directly, we can break it down into a sum of easier polynomial terms. The more terms we add, the more accurate the approximation becomes. And it is important in Computer Science because computers can't calculate functions like $\sin(x)$. Taylor Series solves this problem by breaking down these difficult functions into simple arithmetic operations (like addition and multiplication). Computer can do arithmetic operations quickly and efficiently so it is easier this way.

2.

$$\sin(x) = \sum (-1)^n x^{(2n+1)} / (2n+1)!$$

$$\cos(x) = \sum (-1)^n x^{(2n)} / (2n)!$$

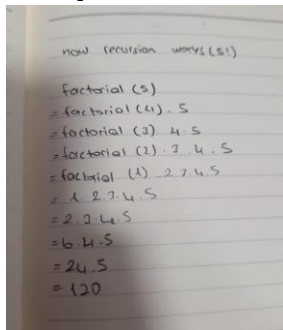
We can pick how many terms based on the accuracy we need. Adding more terms gets even closer to the true value. 5-10 terms are generally chosen, enough to be very accurate in unwasted time.

3.

```
function factorial(n):  
    if n == 0 or n == 1:  
        return 1  
    else:  
        return n * factorial(n-1)
```

The function calls itself with a smaller number until it reaches the base case (0 or 1), then multiplies the results back up.

Example:



4.

```
void calculate_trigonometric_functions(double x_rad)  
{  
  
    double sin_value = sin(x_rad);  
  
    double cos_value = cos(x_rad);
```

```
printf("X (radian): %.4f\n", x_rad);  
  
printf("sin(X):  %.15f\n", sin_value);  
  
printf("cos(X):  %.15f\n", cos_value);
```

```
}
```

$\sin(x)$ and $\cos(x)$ can be used directly with angle in radians as input and a value between -1 and 1 as output.

math.h is very fast, Taylor Series is slow, and Recursive is very slow.

math.h provides perfect accuracy, while both Taylor Series and Recursive methods offer less good accuracy.

math.h requires only 1 line of code, Taylor Series needs many lines, and Recursive needs more than 1 line but not as much as Taylor Series does.

math.h is really excellent for practical applications, Taylor Series is useful for learning purposes, and Recursive is suitable for education only.